PILE & WELL FOUNDATIONS

1. Pile Foundation – Introduction

Pile = slender structural member (timber / concrete / steel) that transfers load to deeper, stronger strata by skin friction + end bearing.

When are piles used?

- Top soil weak / compressible, firm stratum at depth (> 3–4 m)
- Heavy concentrated loads – bridges, towers, silos, jetties
- Uplift & lateral loads (chimneys, transmission towers, quay walls)
- Scour-prone river beds, expansive / collapsible soils
- High water table – deep excavation uneconomical

2. Classification of Piles

(a) Based on function / load transfer

- **End-bearing (point-bearing) pile** – tip rests on rock / hard stratum; load mostly by tip: $Q_u \approx Q_p$
- **Friction pile** – floats in soft clay / loose sand; load by shaft: $Q_u \approx Q_s$
- **Combined** – $Q_u = Q_p + Q_s$ (most common)
- **Compaction pile** – densify loose granular soil
- **Tension / uplift pile** – resist upward pull (towers, basements)
- **Batter (raker) pile** – inclined; resist horizontal loads
- **Sheet pile** – retaining walls / cofferdams
- **Anchor pile** – hold down bulkheads; **Fender pile** – protect jetties

(b) Based on material

- Timber, Concrete (precast / cast-in-situ / prestressed), Steel (H, pipe), Composite

(c) Based on method of installation

- **Driven** (displacement) – hammer / vibro; **Bored** (non-displacement); **Driven cast-in-situ**; **Screw piles**; **Jacked**

(d) Based on size

- Small dia. < 600 mm; Large dia. ≥ 600 mm (bored piers / caissons)

3. Types of Piles – Material wise

Type	Features / Uses	Merits & Demerits
Timber	Sal, teak, deodar; 6–12 m long, 0.2–0.5 m dia. Temporary, marine fenders, light loads (≤ 250 kN)	Cheap, light, easy handling. Rots above W.T.; marine borer attack; low capacity
Precast RC	Square / octagonal; cast in yard, cured, then driven. Reinforced for handling. 300–600 mm; ≤ 30 m	Durable, good quality control. Heavy, hard to cut / extend; need driving equipment
Prestressed	Long, slender; resists driving tension & bending	Light, high capacity, less cracking; costly
Cast-in-situ	Bored / driven with casing – Franki, Raymond, Vibro, Under-reamed	Any length, no handling stress. Necking / segregation risk
Steel	H-pile, pipe pile (filled with concrete). Easy to splice	High capacity, deep penetration. Corrosion, costly
Composite	Timber + concrete / steel + concrete (upper part durable)	Economical for long piles with high W.T.

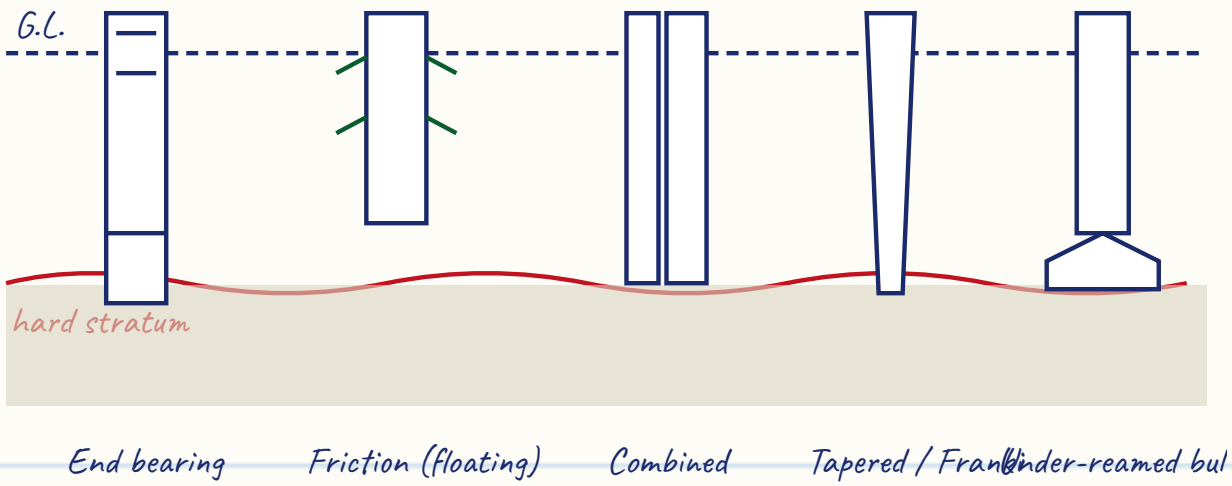


Fig 1: Pile types by load transfer & shape

Under-reamed pile (IS 2911 Pt 3): bulb dia = 2–3 × shaft; great in expansive (black cotton) soils.

5. Load Transfer Mechanism

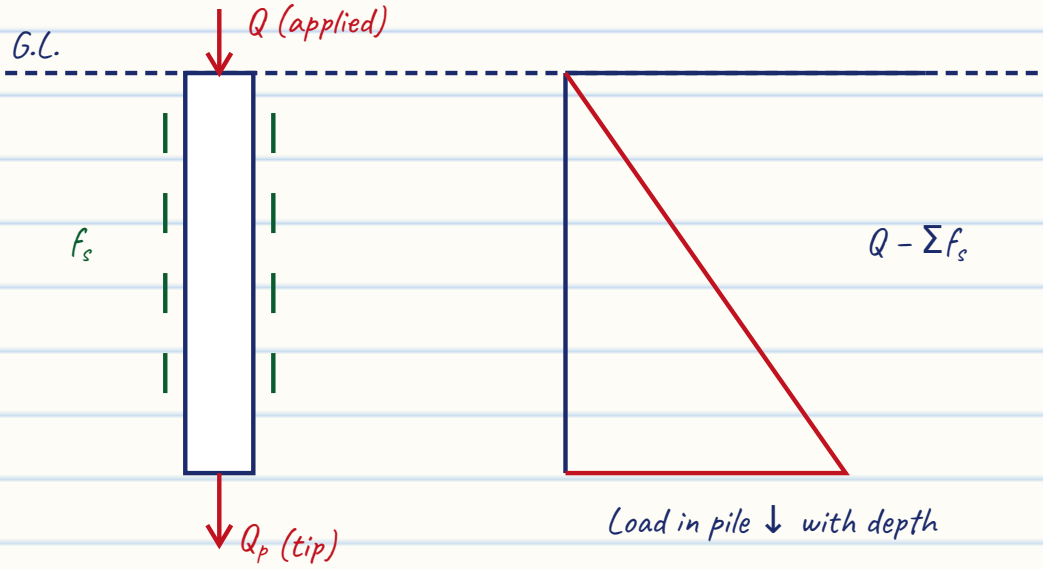


Fig 2: Shaft friction mobilises first (≈ 0.5–1 % d settlement); tip needs ≈ 10 % d

$$Q_u = Q_p + Q_s \quad ; \quad Q_a = Q_u / F.S.\ (2.5)$$

6. Load Capacity – Static Analysis

$$Q_u = q_p A_p + \Sigma f_s A_s - W_p(small)$$

A_p = tip area; A_s = perimeter × length; q_p = unit end bearing; f_s = unit skin friction.

(A) Piles in Clay ($\varphi = 0$; undrained)

End bearing:

$$Q_p = N_c\, c_u\, A_p \ , \ N_c = 9 \ (L/d > 4)$$

Skin friction – α method (total stress):

$$f_s = \alpha\, c_u \ \Rightarrow \ Q_s = \alpha\, c_u\, (\pi\, d\, L)$$

c_u (kPa)	≤ 25	25–50	50–75	> 75
α (typical)	1.0	0.8	0.6	0.4–0.5

- α = adhesion factor = adhesion / cohesion; ↓ as c_u ↑ (remoulding, water film)
- Layered clay: $Q_s = \Sigma\, \alpha_i\, c_{ui}\, P\, \Delta L_i$

λ method (Vijayvergiya & Focht) – driven piles in clay:

$$f_{av} = \lambda\, (\bar{\sigma}_v + 2\, c_u)$$

- $\bar{\sigma}_v$ = mean effective vertical stress along shaft; c_u = mean undrained strength
- λ depends on pile length (passive pressure effect):

L (m)	5	10	20	30	40
λ	0.34	0.25	0.17	0.14	0.13

β method (effective stress – NC clay long term):

$$f_s = \beta \sigma'_v, \beta = K \tan\delta \approx (1 - \sin\varphi') \tan\varphi' \approx 0.25 - 0.35$$

α → short-term (undrained). β → long-term / drained check. Use smaller of the two for design.

(B) Piles in Sand (c = 0 ; drained)

End bearing (Meyerhof):

$$q_p = \sigma'_v N_q \leq q_l = 0.5 P_a N_q \tan\varphi$$

- σ'_v = effective overburden at tip (limited to critical depth $z_c \approx 15\,d$ for loose–medium, 20 d dense sand)
- N_q from Meyerhof's chart (depends on φ and L/d; φ increased for driven piles: $\varphi = (\varphi_1 + 40)/2$)
- P_a = 100 kPa; limiting value q_l (the pile tip "critical depth" concept)

From SPT (Meyerhof):

$$q_p = 40\,\bar{N}(L_b/d) \leq 400\,\bar{N}\,(kPa)$$

$$f_s = 2\,\bar{N}(\text{driven}), \quad f_s = 1\,\bar{N}(\text{bored}) \quad (kPa)$$

Skin friction – β (effective stress) method:

$$f_s = K\sigma'_v \tan\delta \quad (\sigma'_v \text{ constant below } z_c)$$

Pile	K (loose → dense)	δ
Driven, displacement	1.0 – 2.0 ($K_0 = 1 - \sin\varphi$ to 1.75 K_0)	0.75 φ
Bored	0.7 – 1.0	2/3 φ
Steel H / pipe	0.5 – 1.0	20° (steel)

IS 2911 (Part 1/Sec 2) general formula:

$$Q_u = A_p(\tfrac{1}{2}\, d\gamma N_\gamma + \sigma'_v N_q) + \Sigma K\sigma'_{vi} \tan\delta\, A_{si}$$

Rock socketed pile:

- $q_p \approx 3\, q_{u,rock} \sqrt{(1 + 300\, t_d/s_d)}$ (Rowe-Armitage – brief) ; $f_s \approx 0.45 \sqrt{q_u}$ (MPa)

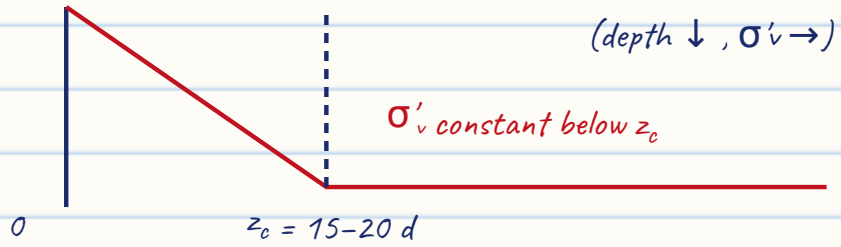


Fig 3: Critical depth – σ'_v and f_s cap

7. Worked Examples – Static Capacity

Example 1 (Clay)

A 400 mm dia. bored pile, 15 m long, in homogeneous clay c_u = 60 kPa, α = 0.6. Find safe load, F.S. = 2.5.

Solution:

$A_p = \pi/4 \times 0.4^2 = 0.1257\ m^2$; $A_s = \pi \times 0.4 \times 15 = 18.85\ m^2$

$Q_p = 9 \times 60 \times 0.1257 = \mathbf{67.9\ kN}$

$Q_s = \alpha\, c_u\, A_s = 0.6 \times 60 \times 18.85 = \mathbf{678.6\ kN}$

$Q_u = 67.9 + 678.6 = 746.5\ kN$

$$Q_a = 746.5 / 2.5 \approx 298.6\ kN \approx 300\ kN$$

Note: ≈ 91 % of capacity from skin friction → a friction pile.

Dynamic Pile Formulae

8. Dynamic Formulae

Idea: Energy delivered by hammer = work done in driving pile into soil + losses. Gives capacity at the time of driving – quick field check for driven piles in granular soils (unreliable in clay: set-up / relaxation).

$$W H \eta_h = Q_u \times S + \text{losses}$$

(i) Engineering News Formula (ENR)

- Drop hammer: $Q_u = W H / (S + 2.5)$ (S in cm, WH in kN·cm), F.S. = 6
- Single-acting steam hammer: C = 2.5 cm ; double-acting: C = 0.25 cm

$$Q_a = W H / [6 (S + C)]$$

W = weight of hammer (kN), H = height of fall (cm), S = final set per blow (cm), C = 2.5 cm (drop).

(ii) Hiley Formula (considers hammer efficiency, restitution, temporary compression)

$$Q_u = [\eta_h W H / (S + C/2)] \cdot [(W + e^2 P) / (W + P)]$$

- η_h = hammer efficiency (0.75–0.85 drop; 0.8–0.9 steam; 0.8 diesel)
- P = weight of pile + helmet + cap ; W = hammer weight
- e = coefficient of restitution : 0.25 (wood), 0.4 (conc. with cushion), 0.5 (steel on steel)
- $C = C_1 + C_2 + C_3$ = temporary elastic compression of (cap + pile + soil)
- Typical: C_1 (cap) 5–10 mm, C_2 (pile) 2–5 mm, C_3 (soil, quake) 0–5 mm … F.S. = 2–3

(iii) Modified Hiley (IS 2911 Part 1) – uses efficiency of blow:

$$Q_u = \eta_h \cdot \eta_b \cdot W H / (S + C/2)$$

$\eta_b = (W + e^2 P) / (W + P)$ = efficiency of blow; if $P > W e^2$, η_b falls → heavy pile needs heavy hammer.

(iv) Danish Formula

$$Q_u = \eta_h E / [S + \sqrt{(\eta_h E L / 2 A E_p)}]$$

- $E = W H$ (hammer energy), L = pile length, A = cross-section area, E_p = Young's modulus of pile
- Second term in denominator = elastic compression of pile (accounts for pile length) ; F.S. = 3

Example 3 (Dynamic formulae)

A precast RC pile (0.3 m × 0.3 m, 12 m, 25 kN/m^3 ⇒ weight P = 27 kN) is driven by a drop hammer of W = 25 kN, fall H = 2.5 m. Average set = 15 mm / blow. Find (a) allowable load by ENR; (b) Hiley: η_h = 0.8, e = 0.4, C = 12 mm (F.S. 2.5).

(a) ENR (drop hammer)

$W H$ = 25 × 250 cm = 6250 kN·cm ; S = 1.5 cm ; C = 2.5 cm

$$Q_a = 6250 / [6 (1.5 + 2.5)] = 6250 / 24 \approx 260 \text{ kN}$$

(b) Hiley

$\eta_b = (W + e^2 P)/(W + P) = (25 + 0.16 \times 27)/(25 + 27) = 29.32 / 52 = 0.564$

$W H \eta_h$ = 0.8 × 25 × 2.5 = 50 kN·m = 50 000 kN·mm (units of S in mm)

$S + C/2$ = 15 + 6 = 21 mm

$Q_u = (50\,000 / 21) \times 0.564 = 2381 \times 0.564 = 1343$ kN

$$Q_a = 1343 / 2.5 \approx 537 \text{ kN}$$

ENR is very conservative (F.S. 6) – gives lower value; Hiley is closer to the load-test results.

9. Pile Load Test (IS 2911 Part 4)

Direct, most reliable method – conducted on **initial test pile** (to establish capacity, test to 2 × design load) and **routine test pile** (on working pile, 1.5 × design load).

Types

1. **Maintained Load Test (MLT / ML)** – load applied in increments; settlement observed
2. **Constant Rate of Penetration (CRP)** – pile pushed at fixed rate; load recorded
3. **Cyclic Load Test** – loading–unloading cycles to separate elastic & plastic settlement

Set-up

- **Kentledge** (dead weight ≥ 1.25× max. test load on a platform) or **reaction piles** with anchored beam (≥ 3 d clear spacing)
- Hydraulic jack + calibrated pressure gauge/load cell; dial gauges (0.01 mm) on independent reference beam

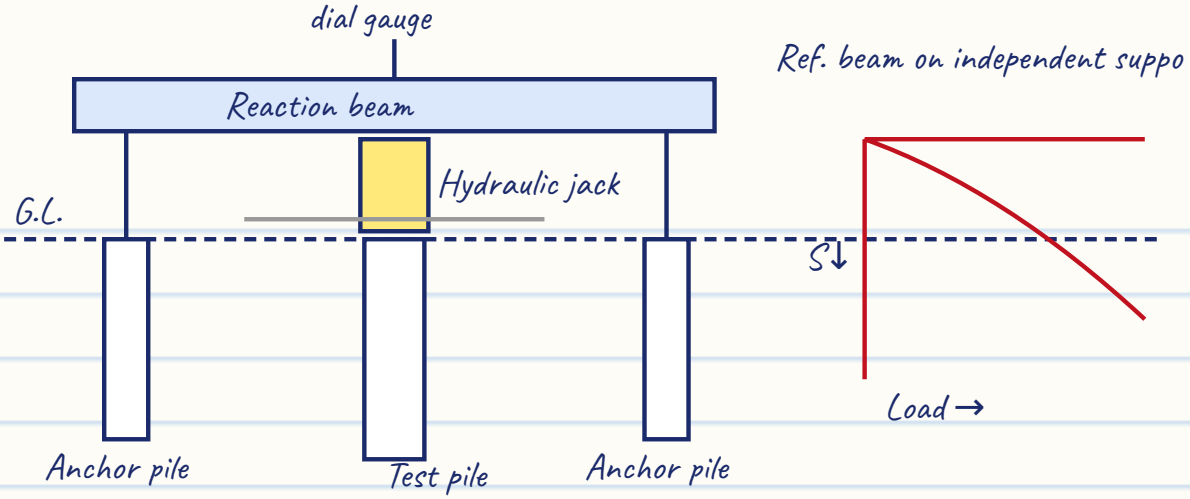


Fig 4: Pile load test by reaction piles; typical load–settlement curve (right)

Maintained load test – procedure

- Apply load in **equal increments = 20 % of safe load**; hold each until rate of settlement ≤ 0.1 mm/h (min. 1 h; or 2 h)

- Continue to 2 × design load (or failure); hold final load 24 h; then unload in same stages
- Read settlement at 5, 10, 20, 30, 60 min … each stage

CRP test

- Jack forces pile at constant rate: **0.75 mm/min (sand), 0.25 mm/min (clay)**; load measured continuously
- Ends at ultimate load or (settlement = 10–15 % d); quicker – hours not days

Interpretation – allowable load (IS 2911 Pt 4) = least of

1. ⅔ of load causing **12 mm** total settlement
2. ½ of load causing **10 % of pile dia.** settlement
3. ⅔ of load for 6 mm net (after rebound) settlement (some cases)

Plot load–settlement; ultimate load by tangent/ double-tangent intersection or Davisson offset (0.15 + d/120 in.).

Pile Groups

10. Pile Groups

Piles usually used in **groups** (3–9 piles) connected by a **pile cap**. Capacity of group ≠ N × single pile because stress zones overlap.

$$\text{Group efficiency } \eta_g = Q_{g(u)} / (n\,Q_u)$$

Spacing

- Min. c/c = 2.5 d (IS: 2.5 d end-bearing; 3 d friction). Typical 3 d
- Edge distance ≥ 1.5 d from pile centre to cap edge

Piles in Sand

- Driven piles densify sand → η_g may be > 1 (spacing 3–4 d). For bored piles $\eta \approx$ 2/3–1.
- Design: $Q_g = n\,Q_u$ (η = 1, for $s \geq 3d$) ; settlement governs

Piles in Clay

- Capacity = **lesser** of (i) $n \times Q_u \times \eta$ and (ii) **block failure** of soil + piles acting as an equivalent pier

$$Q_{g(u)} = 2\,(L_g + B_g)\,D\,\bar{c}_u + N_c\,c_u\,L_g\,B_g$$

L_g, B_g = overall plan dimensions of group (out-to-out) ; D = pile length ; N_c from Skempton ($\approx$ 9, $5(1+0.2B/L)(1+0.2D/B)$).

Converse–Labarre formula (n_1 rows × n_2 columns)

$$\eta_g = 1 - \theta\,[(n_1-1)\,n_2 + (n_2-1)\,n_1] / (90\,n_1\,n_2)$$

$\theta = \tan^{-1}(d/s)$ in degrees; d = pile diameter, s = c/c spacing.

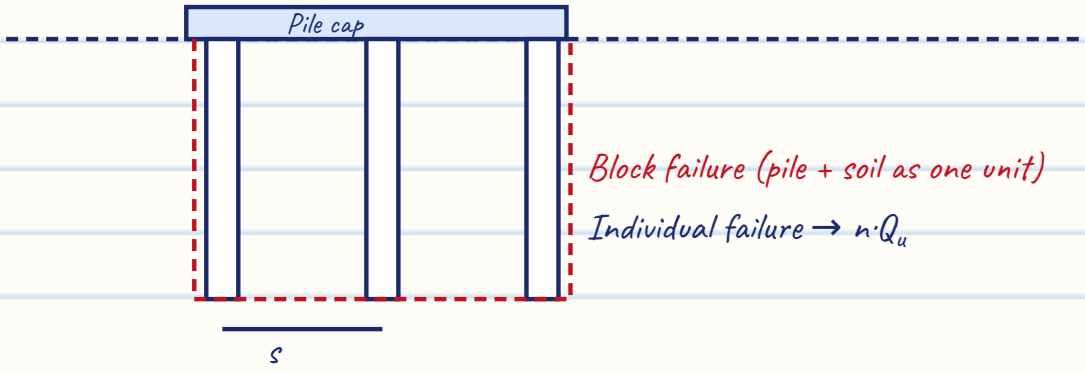


Fig 5: Block failure of pile group in clay

Example 4 – Group capacity

A 3 × 3 group of 400 mm dia. piles, 12 m long, c/c spacing 1.2 m, in clay c_u = 50 kPa, α = 0.7. Find group capacity (F.S. = 2.5).

Single pile: $Q_p = 9 \times 50 \times 0.1257 = 56.5$; $Q_s = 0.7 \times 50 \times \pi \times 0.4 \times 12 = 527.8$; Q_u = **584 kN**

(i) Individual action (Converse–Labarre):

$\theta = \tan^{-1}(0.4/1.2) = 18.43°$; $\eta = 1 - (18.43/90)(2\times3 + 2\times3)/9 = 1 - 0.2048 \times 1.333$ = **0.727**

$Q_g = 0.727 \times 9 \times 584 =$ **3821 kN**

(ii) Block failure: $B_g = L_g = 2 \times 1.2 + 0.4 = 2.8$ m

Perimeter shear = $4 \times 2.8 \times 12 \times 50 = 6720$ kN ; base = $9 \times 50 \times 2.8^2 = 3528$ kN

Q_g = 10 248 kN ($\gg$ 3821)

Governing $Q_{g(u)}$ = 3821 kN $\Rightarrow$ Q_a = 3821/2.5 $\approx$ 1528 kN

II. Settlement of Pile Groups

- Single pile settlement small; group settles more – deeper stress bulb ($\approx$ 1.5 B_g)
- **Equivalent raft method (clay)** : load acts on imaginary raft at **2L/3** (friction piles) or at pile tip (end-bearing) ; spread by **2 V : 1 H** ; compute consolidation settlement

$$S_g = \Sigma\,[C_c\,H / (1+e_0)]\,\log[(\sigma'_0 + \Delta\sigma')/\sigma'_0]$$

- **Sand (Vesic):** $S_g = S\sqrt{(B_g / d)}$ (S = single pile settlement)
- **Meyerhof (SPT):** S_g (mm) $= 0.96\, q\sqrt{(B_g I_c)} / N$; $I_c = 1 - L/(8B_g) \geq 0.5$
- Permissible: isolated 12–25 mm (IS); differential 1 in 500.

12. Negative Skin Friction (Down-drag)

Soft layer / fill settling faster than the pile ⇒ soil drags pile **downward** – extra load!

- **Causes:** fill on soft clay, lowering of water table, consolidating clay, collapse of soil
- Neutral plane: level where soil & pile settle equally

$$Q_n = P \cdot L_n \cdot f_n ; f_n = c_u \text{ (clay) or } \beta\, \sigma'_v \text{ (sand)}$$

- Design: $Q_a = (Q_u - Q_n) / F.S.$ (friction piles) ; for end-bearing: pile structural load = $Q + Q_n$
- **Reduce:** bitumen coating, sleeves, larger preload, increase pile dia. (less area/perimeter)

Well Foundations

13. Well (Caisson) Foundation – Introduction

A **well foundation** is a hollow masonry / RC box (circular, rectangular ...) sunk by excavating soil inside, then plugged and filled – used for **bridge piers & abutments** in rivers; depth 20–40 m or more. India: typical for large rivers (Ganga, Yamuna, Godavari).

Disadvantages

- Tilts & shifts; difficult to sink through boulders / clay (hang-up); needs skilled supervision

14. Types of Well (Caisson) Foundations

1. **Open caisson (open well)** – open at top & bottom; sunk by dredging through the open; most common for bridges. Variants: Well/block, pier-well, box.
2. **Box caisson** – closed at bottom, open at top; prefabricated on land, floated, sunk on prepared bed (levelled, firm soil, water depth ≤ 12 m; docks, jetties)
3. **Pneumatic caisson** – closed top; **compressed air** keeps water out in working chamber; for very deep (> 30 m) bed, no dredging; caisson disease for workers; costly; max. ~ 35 m (3.5 atm)
4. **Floating caisson, monolith** (multi-dredge-well)

Type	Open	Box	Pneumatic
Bottom	Open	Closed	Working chamber
Top	Open	Open	Closed (air lock)
Depth	Large (≤ 40 m)	≤ 12 m	≤ 35 m
Use	Bridge piers	Harbours	Deep, clay/boulders

15. Shapes of Wells

- **Circular** – most common: uniform stresses, least perimeter, easy sinking, minimum tilt, good for river currents from any angle
- **Double-D** – oblong piers, two dredge holes; **Twin circular** – wide piers/ abutments
- **Double octagonal** – 2 dredge holes (aligned with flow)
- **Rectangular / Square** – large, multiple dredge holes; need corner stiffening

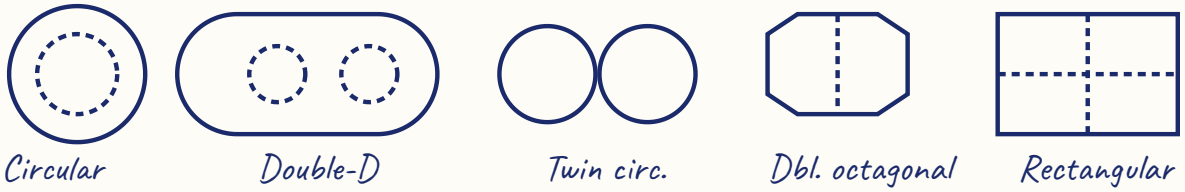


Fig 6: Plan shapes (dotted = dredge holes)

16. Components of a Well Foundation

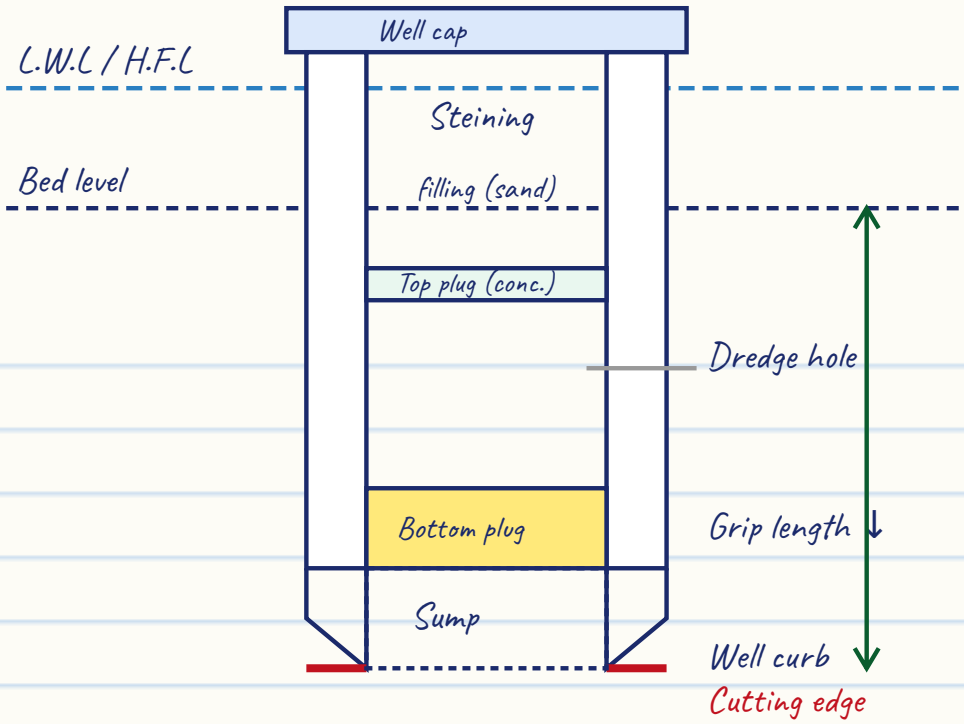


Fig 7: Section through an open-type well

Component	Function / Detail
Cutting edge	Steel angle / plates at bottom; cuts soil; MS, 20 mm+ thick
Well curb	RC wedge above cutting edge, inner face 30°–37° to horizontal; strong; M25 conc.
Steining	Well wall; brick / stone masonry or RCC; min. 500 mm; heavy enough to sink & resist skin friction
Bottom plug	Concrete (tremie) filling under water, 0.6–1 m above curb top ; forms the seal
Sand filling	Fills the well between plugs – economical; gives weight
Top plug	≥ 300 mm plain concrete under well cap
Well cap	RC slab at top; distributes pier loads ; ≥ 750 mm thick
Sump	Extra excavation below cutting edge ≥ 0.5 m (1.5 m max) ; allows seating

17. Functions of Well Foundations

- Transfer large vertical loads (via base + skin friction) to sound strata
- Resist lateral loads (passive earth pressure on sides) & overturning
- Provide grip length below scour level – safeguard against scour
- Act as a cofferdam / retaining structure to excavate in water
- Resist impact of floating debris, boats; give mass for pier / abutment

18. Forces on Well Foundations (IRC 6 / 78)

Force	Details
Dead load	Superstructure + pier + cap + well (buoyancy reduces)
Live load + impact	Vehicular load, impact factor (IRC Class A/70R)
Wind	On superstructure, live load, exposed pier
Water current	$P = 52\,K\,V^2$ (kg/m^2), K = 1.0 square, 0.66 circular piers; V = velocity (m/s) at the max scour...(varies with depth)
Tractive / braking	Longitudinal horizontal force
Earth pressure	For abutment wells: Rankine / Coulomb
Buoyancy	Upward = weight of displaced water (full for soil less than 1m...)
Seismic, temperature, shrinkage, centrifugal	As per IRC 6

Scour (Lacey)

$$d_{sm} = 0.473\,(Q/f)^{1/3} \;;\; \text{max. scour} = 2\,d_{sm} \text{ (at pier nose)}$$

- f = Lacey silt factor = $1.76\sqrt{d_{mm}}$; Q = design discharge (m^3/s)
- Founding level = max. scour + **grip length** (≥ 1/3 × max scour depth below M.S.L.); in rock ≥ 0.6 m socket

Stability checks

- Base pressure ≤ safe bearing capacity ($\sigma_{max} = P/A + M/Z$)
- Lateral stability – **Terzaghi / IRC** method: passive resistance of soil mobilised on side faces; $\sigma \le$ allowable passive pressure ÷ F.S. (2)
- Tension in steining/ well cap; check in-flow & sliding

Max/min base pressure: tension never allowed at base (kern); IRC permits 25 % more stress with wind/seismic.

19. Sinking of Wells

Well sunk under its own weight by **excavating soil from inside** (dredge hole); steining is built in **lifts of 1.5–2 m**, each allowed to set before sinking.

1. **Preparation:** in water → build **sand island** / cofferdam / floating caisson; on land → level bed
2. **Laying the cutting edge & curb** on prepared platform; centre accurately; build first lift of steining
3. **Dredging** with **grab (clamshell)**, dredge-hole crane or divers; evenly from all dredge holes
4. Well sinks when W_{well} + kentledge > skin friction + end resistance (check each stage)
5. Repeat lifts, check **tilt (plumb)** and **shift** each shift
6. At founding level: inspect strata, clean, place **bottom plug** by tremie, fill, **top plug**, cap

Aids to sinking

- **Kentledge** (extra load – rails, sand bags, concrete blocks)
- **Water jetting** (outside) / **bentonite lubrication** – reduces skin friction
- **Blasting** (minor, with caution) for hard strata / boulders ; air-jets
- Dewatering (bailing) to allow dry dredging (reduce buoyancy)
- Pneumatic cavity for obstruction removal

Difficulties

- **Hang-up** – friction ≫ weight; **sudden sinking / blow** in quicksand; obstructions (tree trunks, boulders); skin friction ↑ in stiff clay

20. Tilts and Shifts

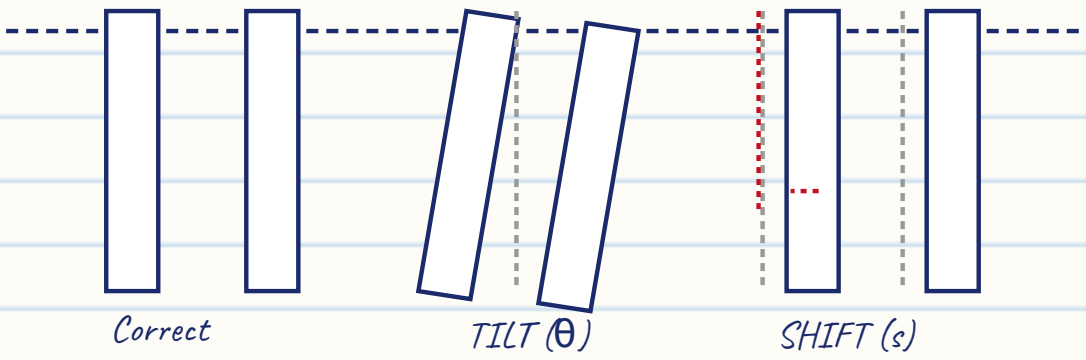


Fig 8: Tilt = angular deviation from vertical; Shift = lateral displacement of centre

- **Tilt** = H : V deviation (e.g. 1 in 100) ; **Shift** = horizontal displacement of centre at base. IRC limits: tilt ≤ 1 in 80 (during sinking: 1 in 100), shift ≤ 150 mm

Causes of Tilts & Shifts

- **Uneven strata** – hard patch / sloping rock on one side, boulders, tree trunks
- Unequal dredging among dredge holes; sudden collapse of soil / blow
- Eccentric kentledge; non-uniform skin friction; unequal steining weight
- Scour on one side; current / wind pressure; imperfect first lift (curb not horizontal)
- Sudden release of friction while sinking

Remedial measures – Tilt

- **Eccentric dredging** – dredge more from the high side (side opposite to tilt direction), leave soil on low side
- **Eccentric loading** (kentledge) on the high side
- **Water jetting / pulling** with wire ropes + winches (anchored); strutting on low side with piles/ sandbags
- **Obstruction removal** by divers, blasting or chiselling (chisel bar)
- Reduce skin friction on the high side; dewatering

Remedial measures – Shift

- Apply **horizontal pull** with ropes, tackle; **strutting** / anchor against the shifted side
- Induce tilt deliberately opposite then correct; dredge / water jet on the side to which the well should move

- **Prevention:** accurate setting, sand island level, **guide frames**, check every 1 m sink with plumb bob / theodolite

Tilt (1 in n) = (shift of base centre – shift at top) / depth sunk

Total shift ≈ tilt × depth + original shift. Tilt corrected gradually – never rush.

Important Questions (Unit V)

1. Classify piles; explain load-carrying mechanism with sketches. (Types: friction/end-bearing)
2. Derive load capacity of a pile in clay (α method) and sand. **(Num.)**
3. Explain Hiley's formula; compare with ENR & Danish. Mention the limitations of dynamic formulae.
4. Describe the pile load test; how is the safe load determined? CRP v/s maintained load test.
5. What is group efficiency? Converse–Labarre; block failure of pile group in clay. **(Num.)**
6. Settlement of a pile group; equivalent raft method. Negative skin friction – causes & remedies.
7. Draw a neat sketch of a well foundation, label components, functions of each.
8. Types and shapes of wells; forces acting; how is a well sunk?
9. Causes of tilts & shifts and remedies – short note.
10. Pneumatic caisson vs open caisson; scour depth & grip length.